

Question-1 $\Rightarrow$ Brownian Motion paths and Quadratic Variation

(a)

Ans few defining properties of standard Brownian Motion

$\rightarrow$ A process $\{W_t\}_{t \geq 0}$ is a standard Brownian Motion if:

(i) $W_0 = 0$

$\rightarrow$ starts at zero

$\rightarrow$ Financial meaning: $\rightarrow$ asset prices are measured relative to the current time.

(ii) Independent Increments

$\rightarrow$ For $0 \leq t_0 < t_1 < \dots < t_n$

$W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}$ are Independent.

$\rightarrow$ Financial meaning: $\rightarrow$ future price changes do not depend on the past changes.

(iii) Gaussian increments

$\rightarrow$ For $s < t$,

$$W_t - W_s \sim N(0, t-s)$$

$\rightarrow$ Financial meaning: $\rightarrow$ returns over a time interval are normally distributed.

(iv) Continuous paths

$\rightarrow t \mapsto W_t$ is continuous almost surely.

$\rightarrow$ Financial meaning: $\rightarrow$ prices evolve continuously without jumps.

(b)

Ans

Assume $s < t$,

$$\text{then } W_t = W_s + (W_t - W_s)$$

Then,
$$E[W_s W_t] = E[W_s (W_s + (W_t - W_s))] \\ = E[W_s^2] + E[W_s (W_t - W_s)]$$

since $W_t - W_s$ is independent of W_s ,

$$E[W_s (W_t - W_s)] = E[W_s] E[W_t - W_s] = 0$$

Hence,
$$E[W_s W_t] = E[W_s^2] = s.$$

since $E[W_s] = E[W_t] = 0$,

$$\text{cov}(W_s, W_t) = s$$

$\therefore \text{Cov}(W_s, W_t) = \min(s, t).$

(c)

Ans

partition :

$$0 = t_0 < t_1 \dots < t_n = T$$

define :

$$Q^n = \sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2$$

let $A_k = t_{k+1} - t_k$

since, $W_{t_{k+1}} - W_{t_k} \sim N(0, A_k)$

$$E[(W_{t_{k+1}} - W_{t_k})^2] = A_k$$

Thus,
$$E[Q^n] = \sum_k A_k = T$$

$\therefore E[Q^n] = T$

(d)

Ans

for $X \sim N(0, \sigma^2)$ $E[X^4] = 3\sigma^4$

Hence, $\text{Var}(X^2) = 3\sigma^4 - \sigma^4 = 2\sigma^4$

$$\circ \circ \quad \text{Var}(Q^n) = \sum_k 2(\Delta_k)^2$$

Now,

$$\sum_k (\Delta_k)^2 \leq \|\pi\| \sum_k \Delta_k = \tau \|\pi\|,$$

$$\text{where } \|\pi\| = \max_k \Delta_k$$

$$\text{so, } \text{Var}(Q^n) \leq 2\tau \|\pi\|$$

$$\text{As } \|\pi\| \rightarrow 0, \quad \text{Var}(Q^n) \rightarrow 0$$

$$\text{since, } E[Q^n] = \tau$$

$$E[(Q^n - \tau)^2] = \text{Var}(Q^n) \rightarrow 0$$

$$\text{Hence, } Q^n \rightarrow \tau \text{ in } L^2$$

Question-2 Itô's Lemma and Geometric Brownian Motion

(a)

Ans

$$\text{If } dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t$$

and $f(t, x)$ is twice differentiable in x and once in t , then,

$$df = \left(f_t + \mu f_x + \frac{1}{2} \sigma^2 f_{xx} \right) dt + \sigma f_x dW_t.$$

The extra term $\frac{1}{2} \sigma^2 f_{xx} dt$ is called Itô correction term.

ordinary chain rule fails because brownian motion has non-zero quadratic variation.

$(dW_t)^2 = dt$. so second-order terms do not disappear.

(b)

Ans

$$\text{let } f(s) = \ln s.$$

Then, $f'(s) = \frac{1}{s}$ $f''(s) = -\frac{1}{s^2}$

using Itô's lemma,

$$d(\ln S_t) = f'(S_t) dS_t + \frac{1}{2} f''(S_t) (dS_t)^2$$

substitute,

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

also, $(dS_t)^2 = \sigma^2 S_t^2 dt.$

$$\therefore d(\ln S_t) = \frac{1}{S} (\mu S dt + \sigma S dW) + \frac{1}{2} \left(-\frac{1}{S^2}\right) (\sigma^2 S^2 dt)$$

Hence, $d(\ln S_t) = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW_t.$

Integrate from 0 to t ,

$$\ln S_t - \ln S_0 = \left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W_t.$$

$$\therefore S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W_t}.$$

(c)

Ans

The term $-\frac{\sigma^2}{2}$ appears because volatility reduces the average log-growth rate.

$$E[S_t] = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right) t} E[e^{\sigma W_t}]$$

since, $W_t \sim N(0, t)$

$$\text{and } E[e^{\sigma W_t}] = e^{\sigma^2 t / 2}.$$

Thus,

$$E[S_t] = S_0 e^{\mu t}$$

Wrong solution $\Rightarrow S_t^{\text{wrong}} = S_0 e^{\mu t + \sigma W_t}$

Then,

$$E[S_t^{\text{wrong}}] = S_0 e^{ut + \frac{\sigma^2 t}{2}}$$

for $\sigma = 0.2$ $t = 10$

$$\therefore \text{error factor} = e^{0.2} \approx 1.221$$

so expected price is overestimated by about 22.1%.

Question 3: Euler-Maruyama simulation

(a)

Ans The OU process is:

$$dX_t = -\theta X_t dt + \sigma dW_t.$$

Hence, drift $\mu(X_t) = -\theta X_t$.

diffusion $\sigma(X_t) = \sigma$.

Financial Interpret.:-

The drift term $-\theta X_t$ always pulls the process towards zero.

- If $X_t > 0$, then $-\theta X_t < 0$, so the drift pushes the process downward.
- If $X_t < 0$ then $-\theta X_t > 0$, so drift pushes the process upward.

(b) For time step Δt ,

$$\begin{aligned} X_{k+1} &= X_k - \theta X_k \Delta t + \sigma \sqrt{\Delta t} Z_k \\ &\left(\text{where } Z_k \sim N(0,1) \right) \\ X_{k+1} &= (1 - \theta \Delta t) X_k + \sigma \sqrt{\Delta t} Z_k. \end{aligned}$$

For stability, $|1 - \theta \Delta t| < 1$
 $\Delta t < 2/\theta$

(c) It means at $t \rightarrow \infty$, the distribution of X_t approaches this normal distribution regardless of starting value X_0 .